

ASSESSMENT TOOL – CONCEPT MAP

CSA0311-DS

Concept Maps: Recursion, Call Stack & Memory Usage

Understanding recursion internals and comparing recursion with iteration for hierarchical, nested data structures

Question 1: Recursion in a Nested File System

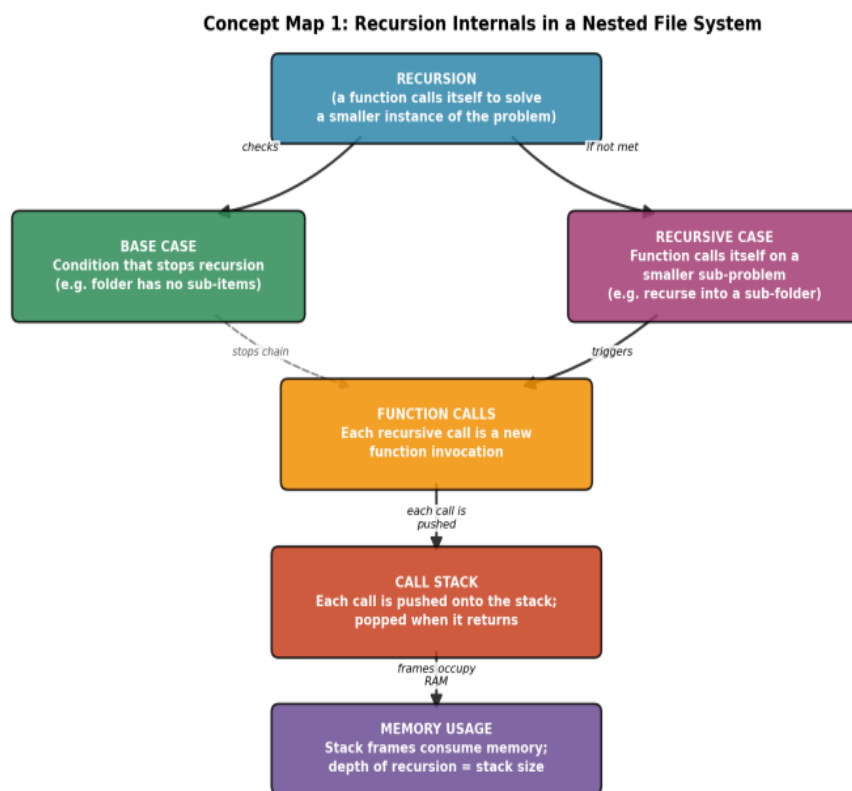


Figure 1. Concept map showing the chain: Recursion → Base Case → Recursive Case → Call Stack → Function Calls → Memory Usage.

Question 2: Recursion vs. Iteration for Nested Data Processing

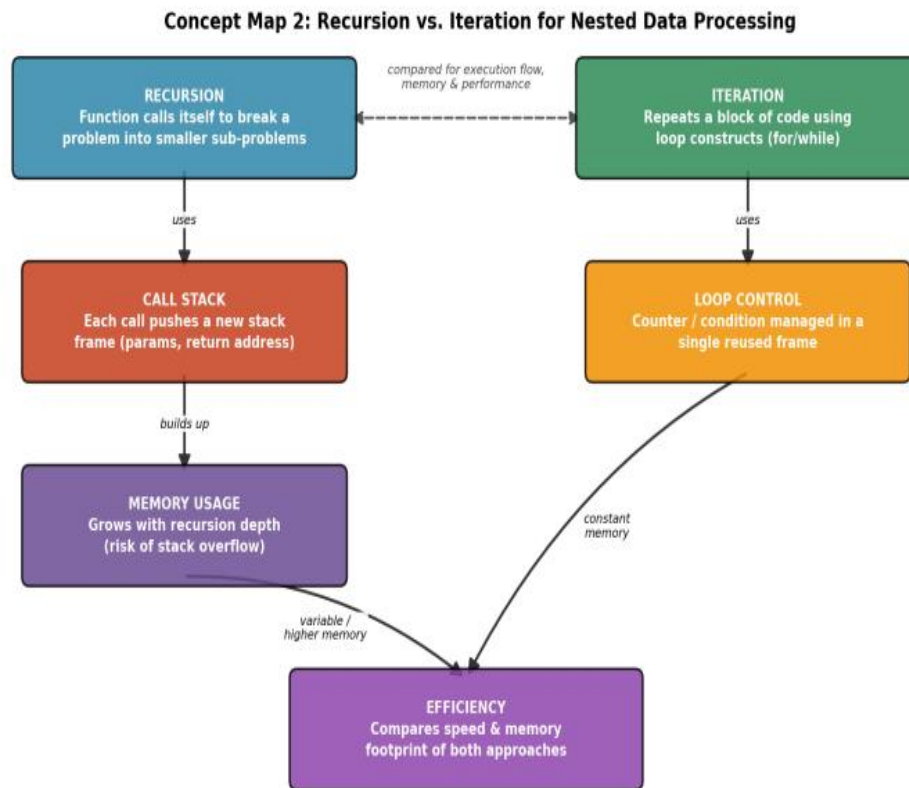


Figure 2. Concept map showing the chain: Recursion → Call Stack → Memory Usage, contrasted with Iteration → Loop Control → Efficiency.